

## SIAULIAI INTL, Lithuania

WMO: 265021

Lat: 55.892N

Lon: 23.397E

Elev: 453

StdP: 14.46

Time Zone: 2.00 (EUE)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -4.4       | 1.9 | -8.6                          | 3.5 | -4.2 | -2.2 | 5.0 | 3.0  | 25.9                  | 31.8 | 23.9 | 31.2 | 4.2                      | 70   | 0.602 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.7                   | 82.8            | 67.1 | 79.6 | 65.6 | 76.8 | 64.0 | 69.9                | 78.3 | 67.8 | 76.2 | 65.9 | 73.7 | 8.6                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.6                            | 99.6 | 73.6 | 64.7 | 92.9 | 71.2 | 62.9 | 87.1 | 69.3 | 34.1          | 78.1 | 32.5 | 76.5 | 31.0 | 73.8 | 77.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |      |
|-------------------|------|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|------|
| 1%                |      | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |      |
| Min               |      | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min   | Max        | Min   | Max        | Min   | Max        |       |      |
| (a)               | (b)  | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |
| 21.7              | 19.2 | 17.3 |                            | DB   | -8.9 | 89.0               | 8.1                                                | 2.7       | -14.8 | 91.0       | -19.5 | 92.6       | -24.1 | 94.1       | -30.0 | 96.1 |
|                   |      |      |                            | WB   | -9.3 | 73.6               | 8.0                                                | 2.1       | -15.1 | 75.1       | -19.7 | 76.3       | -24.2 | 77.5       | -30.0 | 79.0 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 44.8   | 24.1  | 24.7  | 32.3  | 44.4  | 54.3  | 60.1  | 64.7  | 63.7  | 55.2  | 45.0  | 37.5  | 30.7  |
|                                                                                  | DBStd       | 16.18  | 11.39 | 11.18 | 8.72  | 7.24  | 7.24  | 5.94  | 5.45  | 5.18  | 5.71  | 6.66  | 6.96  | 8.77  |
|                                                                                  | HDD50       | 3468   | 801   | 709   | 549   | 199   | 35    | 2     | 0     | 0     | 18    | 182   | 375   | 598   |
|                                                                                  | HDD65       | 7504   | 1266  | 1129  | 1014  | 619   | 341   | 170   | 74    | 87    | 296   | 620   | 825   | 1063  |
|                                                                                  | CDD50       | 1583   | 0     | 0     | 0     | 30    | 167   | 303   | 455   | 425   | 175   | 27    | 1     | 0     |
|                                                                                  | CDD65       | 143    | 0     | 0     | 0     | 0     | 8     | 22    | 64    | 47    | 2     | 0     | 0     | 0     |
|                                                                                  | CDH74       | 1284   | 0     | 0     | 0     | 5     | 92    | 251   | 528   | 386   | 22    | 0     | 0     | 0     |
|                                                                                  | CDH80       | 267    | 0     | 0     | 0     | 0     | 7     | 46    | 127   | 86    | 1     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 8.3    | 9.5   | 8.7   | 8.8   | 8.2   | 7.5   | 7.6   | 7.2   | 7.0   | 7.4   | 8.4   | 9.1   | 10.1  |
| Precipitation                                                                    | PrecAvg     | 23.4   | 1.3   | 1.1   | 1.3   | 1.4   | 1.9   | 2.5   | 3.0   | 2.8   | 2.2   | 2.2   | 2.1   | 1.7   |
|                                                                                  | PrecMax     | 31.3   | 2.4   | 1.8   | 3.1   | 3.9   | 6.7   | 5.7   | 5.7   | 6.4   | 5.0   | 7.2   | 4.4   | 2.8   |
|                                                                                  | PrecMin     | 15.7   | 0.3   | 0.2   | 0.2   | 0.2   | 0.4   | 0.2   | 0.1   | 0.2   | 0.3   | 0.2   | 0.6   | 0.5   |
|                                                                                  | PrecStd     | 3.8    | 0.6   | 0.4   | 0.7   | 0.9   | 1.1   | 1.2   | 1.5   | 1.5   | 0.9   | 1.4   | 0.8   | 0.6   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 46.6  | 44.5  | 57.5  | 73.7  | 80.8  | 85.3  | 87.9  | 87.8  | 77.3  | 64.3  | 52.8  | 48.4  |
|                                                                                  |             | MCWB   | 45.7  | 41.1  | 46.3  | 55.3  | 65.2  | 68.1  | 69.2  | 69.4  | 64.2  | 56.9  | 49.9  | 47.3  |
|                                                                                  | 2%          | DB     | 42.7  | 41.2  | 51.9  | 68.0  | 76.7  | 80.3  | 83.7  | 82.1  | 71.8  | 60.0  | 50.0  | 45.8  |
|                                                                                  |             | MCWB   | 41.4  | 38.9  | 44.0  | 52.1  | 62.4  | 65.3  | 68.1  | 67.2  | 61.7  | 55.4  | 48.0  | 44.5  |
|                                                                                  | 5%          | DB     | 39.6  | 38.8  | 47.4  | 63.0  | 73.0  | 76.8  | 80.2  | 78.5  | 68.2  | 57.1  | 48.1  | 42.8  |
|                                                                                  |             | MCWB   | 38.7  | 37.5  | 42.1  | 50.1  | 59.6  | 63.0  | 66.5  | 65.6  | 59.7  | 53.5  | 46.7  | 41.7  |
|                                                                                  | 10%         | DB     | 37.4  | 36.8  | 43.2  | 58.1  | 68.4  | 73.0  | 76.2  | 74.9  | 64.8  | 54.2  | 46.3  | 40.9  |
|                                                                                  |             | MCWB   | 36.7  | 36.0  | 39.7  | 47.4  | 57.1  | 61.2  | 64.8  | 63.5  | 58.3  | 51.4  | 45.3  | 39.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 45.6  | 41.3  | 48.3  | 57.7  | 67.5  | 70.0  | 73.1  | 72.7  | 66.4  | 59.1  | 50.5  | 47.4  |
|                                                                                  |             | MCDB   | 46.3  | 42.8  | 55.0  | 71.5  | 77.5  | 81.1  | 82.5  | 81.4  | 73.4  | 61.8  | 51.9  | 48.1  |
|                                                                                  | 2%          | WB     | 41.3  | 38.9  | 44.9  | 54.0  | 64.2  | 67.1  | 70.4  | 69.4  | 63.5  | 56.2  | 48.4  | 44.3  |
|                                                                                  |             | MCDB   | 42.0  | 40.4  | 50.3  | 65.1  | 73.8  | 76.7  | 78.6  | 77.5  | 69.1  | 59.1  | 49.3  | 45.2  |
|                                                                                  | 5%          | WB     | 38.7  | 37.3  | 42.5  | 51.2  | 61.6  | 64.8  | 68.3  | 67.2  | 61.2  | 54.1  | 46.6  | 41.4  |
|                                                                                  |             | MCDB   | 39.5  | 38.1  | 46.2  | 59.9  | 69.7  | 73.7  | 76.5  | 75.1  | 66.5  | 56.7  | 47.5  | 42.2  |
|                                                                                  | 10%         | WB     | 36.4  | 35.6  | 39.9  | 48.7  | 58.8  | 62.5  | 66.3  | 65.3  | 59.3  | 52.2  | 45.0  | 39.5  |
|                                                                                  |             | MCDB   | 37.1  | 36.2  | 42.9  | 56.4  | 66.1  | 70.6  | 73.8  | 72.5  | 63.9  | 54.4  | 45.9  | 40.3  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.7   | 9.9   | 13.7  | 18.7  | 19.5  | 18.6  | 17.7  | 17.7  | 15.7  | 12.0  | 7.2   | 6.6   |
|                                                                                  |             | MCDBR  | 7.1   | 8.9   | 18.6  | 26.5  | 24.5  | 24.4  | 23.5  | 23.2  | 20.7  | 14.5  | 8.7   | 6.9   |
|                                                                                  |             | MCWBR  | 7.0   | 7.1   | 12.0  | 14.0  | 12.4  | 11.8  | 10.6  | 10.9  | 11.7  | 10.2  | 7.7   | 6.6   |
|                                                                                  | 5% WB       | MCDBR  | 6.9   | 8.4   | 16.8  | 22.5  | 21.4  | 20.9  | 19.7  | 19.8  | 18.0  | 13.3  | 8.1   | 6.7   |
|                                                                                  | MCWBR       | 7.0    | 7.1   | 11.7  | 13.2  | 12.2  | 11.5  | 10.5  | 10.8  | 11.4  | 10.0  | 7.5   | 6.6   |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.290 | 0.321 | 0.354 | 0.393 | 0.383 | 0.376 | 0.409 | 0.405 | 0.369 | 0.346 | 0.329 | 0.289 |
|                                                                                  | taud        |        | 2.200 | 2.190 | 2.230 | 2.231 | 2.313 | 2.360 | 2.287 | 2.300 | 2.382 | 2.405 | 2.276 | 2.204 |
|                                                                                  | Ebn at Noon |        | 193   | 231   | 254   | 261   | 272   | 275   | 262   | 255   | 249   | 224   | 176   | 161   |
|                                                                                  | Edh at Noon |        | 18    | 27    | 33    | 38    | 37    | 36    | 38    | 36    | 29    | 22    | 17    | 14    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 164   | 395   | 811   | 1288  | 1726  | 1792  | 1678  | 1365  | 940   | 460   | 166   | 105   |
|                                                                                  | RadStd      |        | 36    | 78    | 126   | 168   | 151   | 131   | 149   | 141   | 106   | 76    | 23    | 12    |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (24 neighbors) | +0.77 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page